Urban Heat Shade Dash

Find the coolest route through a hot campus

At a glance: Find the coolest route through a hot campus. Plan about 60 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

BY THE END OF THIS ACTIVITY, EACH STUDENT CAN:

- **Explain the core science of this challenge: microclimate, shade, heat stress, urban cooling (Understand).**
- **Design a fair test by controlling one variable while holding others constant (Apply).**
- **Use their own data to decide whether a redesign improved the result (Analyze).**
- **Defend a final claim with specific evidence (Evaluate).**

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- **4 infrared (non-contact) thermometers (plus 2 spares; one per team). Reads the surface temperature it is pointed at, which is what the sun-versus-shade comparison needs; a stick thermometer reads air, not the surface**
- **spare batteries for the IR thermometers (confirm 9V or 2x AAA per model; start with fresh cells and bring 1 spare set)**
- **4 campus maps or route cards (plus 2 spares; one per team), local print**
- **4 clipboards (plus 1 spare)**
- **pencils per team, shared**

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

3 No PPE: this is an outdoor measurement walk with no chemicals, water, or soil. Set up sun and hydration support, run the weather/heat-index check, confirm the buddy system and boundaries, install fresh batteries in the IR thermometers, and brief the rule to point the laser at surfaces only, never at eyes.

01 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Find the coolest route through a hot campus." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Measure and map

Teams walk the route and use the IR thermometer to record surface temperatures at matched sunny and shaded spots, then mark the hottest and coolest surfaces on the campus map.

Phase 6 Route and justify

Each team plots a cool corridor that stays on the coolest measured surfaces and writes the temperature evidence that makes it cooler than the direct path.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Return the IR thermometers, clipboards, and pencils to the bin and collect the maps and score slips; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A team's sun and shade readings look the same. Fix: Confirm the IR thermometer is aimed at the surface (not the sky or the student's shadow) from a steady close distance, that the sunlit spot has been in full sun and the shaded spot is genuinely shaded, and that they read the same surface type in both places; have them retake both readings.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to urban heat and shade.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	**100**

CLEANUP AND SOURCES

Return the IR thermometers, clipboards, and pencils to the bin and collect the maps and score slips. The only consumables are the printed maps and handouts; there are no surfaces to wipe and no batteries to discard unless one died.

SOURCES NASA MyNASAData Urban Heat Island